

KODIAK, AK, USA

WMO: 703500

Lat: 57.751N

Lon: 152.486W

Elev: 80

StdP: 14.65

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 25501

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 9.6	(c) 13.6	(d) -2.3	(e) 4.9	(f) 13.9	(g) 1.2	(h) 5.9	(i) 16.2	(j) 35.9	(k) 33.9	(l) 33.0	(m) 33.8	(n) 11.6	(o) 290	(p) 0.744

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 10.8	(c) 69.9	(d) 57.9	(e) 66.1	(f) 55.9	(g) 63.2	(h) 55.1	(i) 59.4	(j) 67.0	(k) 57.6	(l) 63.3	(m) 56.4	(n) 61.3	(o) 10.0	(p) 310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
56.2	67.4	60.7	54.9	64.4	59.0	53.8	61.7	57.9	25.9	66.9	24.8	63.6	24.0	61.2	68.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 32.2	(b) 27.5	(c) 24.4	(e) 4.2	(f) 76.1	(g) 5.6	(h) 3.6	(i) 0.2	(j) 78.7	(k) -3.1	(l) 80.8	(m) -6.2	(n) 82.8	(o) -10.2	(p) 85.4		
(a) 32.2	(b) 27.5	(c) 24.4	(e) 3.0	(f) 62.6	(g) 5.4	(h) 2.7	(i) -0.9	(j) 64.6	(k) -4.0	(l) 66.2	(m) -7.1	(n) 67.7	(o) -11.0	(p) 69.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.6	31.0	32.3	32.4	38.3	45.0	50.7	55.3	55.9	50.1	41.7	35.0	31.3		
	DBStd	10.81	7.96	6.74	6.63	4.41	4.54	4.60	4.14	3.98	4.25	5.44	6.78	7.52		
	HDD50	3533	589	497	545	352	169	41	2	2	47	260	450	579		
	HDD65	8537	1054	917	1010	802	621	431	304	284	446	724	900	1044		
	CDD50	479	0	0	0	0	13	61	165	186	52	2	0	0		
	CDD65	6	0	0	0	0	0	1	2	3	0	0	0	0		
	CDH74	27	0	0	0	0	0	7	12	8	0	0	0	0		
	CDH80	1	0	0	0	0	0	0	0	1	0	0	0	0		
	WSAvg	10.3	11.7	11.5	11.9	11.1	10.0	8.6	7.4	7.9	9.1	10.5	11.5	12.3		
	PrecAvg	67.8	7.7	5.8	4.7	4.1	4.8	5.5	3.7	4.8	6.8	7.5	6.0	7.2		
(15) Precipitation	PrecMax	87.7	17.2	12.4	9.9	9.3	12.7	16.9	10.2	11.1	12.1	17.1	15.4	19.8		
	PrecMin	38.3	0.2	1.5	0.8	0.3	1.0	0.7	0.8	0.6	1.2	3.1	0.7	1.2		
	PrecStd	13.1	4.4	3.1	2.4	1.9	3.3	3.7	2.2	2.8	2.7	3.7	3.1	4.4		
	PrecStd	13.1	4.4	3.1	2.4	1.9	3.3	3.7	2.2	2.8	2.7	3.7	3.1	4.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.8	44.4	46.7	55.1	65.6	73.3	75.8	74.3	66.1	55.1	49.5	45.5		
		MCWB	40.2	40.4	40.4	44.4	50.6	58.8	61.3	60.8	54.5	48.8	46.0	43.1		
	2%	DB	42.6	42.4	44.1	49.6	60.2	66.4	69.5	69.2	61.8	52.6	47.4	43.9		
		MCWB	41.0	39.7	40.2	42.5	48.3	54.9	58.6	57.9	53.2	47.9	45.2	42.1		
	5%	DB	41.6	41.2	42.4	46.5	55.8	62.2	65.6	65.5	59.1	51.1	46.1	42.4		
		MCWB	40.2	39.1	39.2	40.9	46.6	53.2	56.5	56.2	52.2	47.5	43.8	40.8		
	10%	DB	40.4	39.9	40.7	44.5	52.3	58.4	62.5	62.8	56.8	49.8	44.7	40.9		
		MCWB	39.1	38.0	37.9	40.3	45.4	51.4	55.8	55.4	51.4	46.5	42.3	39.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.1	42.3	43.6	46.2	52.9	59.6	61.9	62.5	57.7	51.6	47.8	44.3		
		MCDB	43.3	43.0	45.0	51.7	62.2	71.9	72.9	71.6	61.6	52.3	49.0	45.0		
	2%	WB	41.7	40.8	41.6	44.0	49.7	56.3	59.4	59.5	55.2	50.3	46.2	42.7		
		MCDB	42.1	41.6	43.0	47.4	57.7	64.6	68.0	65.9	58.9	51.4	47.2	43.5		
	5%	WB	40.6	39.6	40.1	42.4	48.0	53.9	57.5	57.9	53.6	48.5	44.4	41.2		
		MCDB	41.2	40.6	41.6	45.2	54.1	59.9	63.5	63.0	57.0	50.1	45.8	42.2		
	10%	WB	39.3	38.4	38.5	41.3	46.4	52.2	56.3	56.6	52.4	47.3	42.7	39.7		
		MCDB	40.1	39.6	40.2	43.7	51.0	56.9	61.3	61.0	55.4	49.2	44.4	40.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.3	8.8	9.5	9.7	10.2	9.9	9.6	10.8	11.1	10.7	9.5	9.0		
		MCDBR	6.7	8.8	10.6	13.0	17.5	18.3	17.0	15.8	14.7	11.5	8.8	7.4		
		MCWBR	6.2	7.6	8.1	8.9	9.7	9.9	8.8	8.2	8.6	8.0	8.0	7.0		
	5% WB	MCDBR	5.6	7.2	8.3	10.1	14.0	15.2	14.4	13.0	11.3	8.8	7.0	6.4		
		MCWBR	5.8	7.0	7.1	7.7	8.5	8.8	8.3	7.8	7.9	7.9	7.2	6.6		
(40) Clear-Sky Solar Irradiance	taub		0.276	0.320	0.343	0.380	0.387	0.379	0.375	0.366	0.353	0.332	0.303	0.302		
	taud		2.338	2.300	2.265	2.218	2.281	2.398	2.457	2.485	2.464	2.364	2.190	2.052		
	Ebn at Noon		192	227	254	262	268	273	272	266	251	220	165	122		
	Edh at Noon		14	23	31	38	38	35	32	29	25	21	16	14		
(44) All-Sky Solar Radiation	RadAvg		132	311	699	1053	1389	1513	1444	1172	801	436	190	95		
	RadStd		17	45	98	118	152	157	129	127	101	47	27	9		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	+1.39	+2.21	N/A	N/A	N/A	N/A	-355	-459	N/A	N/A
Regional (4 neighbors)	+1.40	+2.39	+2.20	N/A	N/A	N/A	-355	-459	N/A	N/A

Nomenclature: See separate page